EXP NO: 2 UPLOAD AND ANALYZE THE DATA SET GIVEN IN CSV FORMAT AND PERFORM DATA PREPROCESSING AND VISUALIZATION

AIM:

Use sample data set sales-data.csv.

DATASET GIVEN:

Date	Product	Sales	Quantity	Region
01-01-2023	Product A	200	4	North
02-01-2023	Product B	150	3	South
03-01-2023	Product A	220	5	North
04-01-2023	Product C	300	6	East
05-01-2023	Product B	180	4	West
06-01-2023	Product A	210	5	North
07-01-2023	Product C	320	7	East
08-01-2023	Product B	160	3	South
09-01-2023	Product A	230	6	North
10-01-2023	Product C	310	7	East
11-01-2023	Product B	190	4	West
12-01-2023	Product A	240	6	North
13-01-2023	Product C	330	8	East
14-01-2023	Product B	170	3	South
15-01-2023	Product A	250	7	North
16-01-2023	Product C	340	8	East

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("sales_data.xlsx")
print(df.head())
print(df.info())
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True, errors="coerce")
```

```
df["Sales"] = pd.to_numeric(df["Sales"], errors="coerce")
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Sales"] = df["Sales"].fillna(df["Sales"].median())
df["Quantity"] = df["Quantity"].fillna(df["Quantity"].median())
df["Product"] = df["Product"].fillna(df["Product"].mode()[0])
df["Region"] = df["Region"].fillna(df["Region"].mode()[0])
df = df.drop_duplicates()
df = df.dropna(subset=["Date"])
df = df.sort_values("Date")
print("\nCleaned Dataset:")
print(df)
product_sales = df.groupby("Product")["Sales"].sum()
plt.figure(figsize=(7, 7))
plt.pie(
    product_sales,
    labels=product_sales.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Product vs Sales")
plt.show()
region_sales = df.groupby("Region")["Sales"].sum().sort_values()
plt.figure(figsize=(8, 5))
colors = ["#4E79A7", "#F28E2B", "#59A14F", "#B07AA1"]
plt.barh(region_sales.index, region_sales.values, color=colors)
plt.title("Sales by Region")
plt.xlabel("Total Sales")
plt.ylabel("Region")
plt.tight_layout()
plt.show()
monthly_sales = df.groupby(df["Date"].dt.to_period("M"))["Sales"].sum()
plt.figure(figsize=(10, 5))
monthly_sales.plot(kind="line", marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.tight_layout()
plt.show()
```

OUTPUT:

```
        Date    Product  Sales  Quantity Region
0 2023-01-01  Product A    200         4  North
1 2023-01-02  Product B    150         3  South
2 2023-01-03  Product A    220         5  North
3 2023-01-04  Product C    300         6   East
4 2023-01-05  Product B    180         4   West

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Date      16 non-null     datetime64[ns]
 1   Product   16 non-null     object
 2   Sales     16 non-null     int64
 3   Quantity  16 non-null     int64
 4   Region    16 non-null     object
dtypes: datetime64[ns](1), int64(2), object(2)

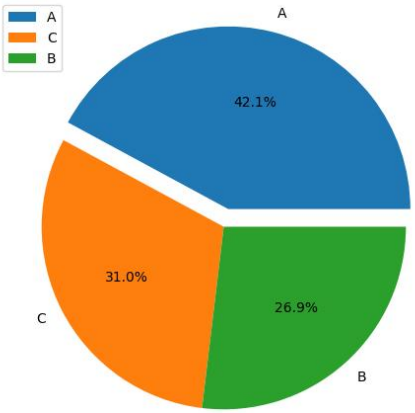
None

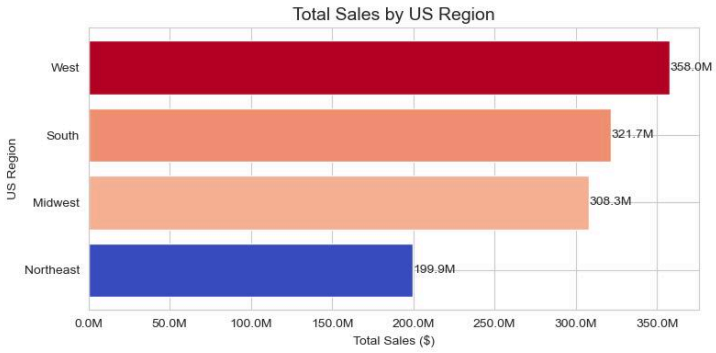
Missing Values:
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64

Duplicate Rows:
0

Cleaned Dataset:
        Date    Product  Sales  Quantity Region
0  2023-01-01  Product A    200         4  North
1  2023-01-02  Product B    150         3  South
2  2023-01-03  Product A    220         5  North
3  2023-01-04  Product C    300         6   East
4  2023-01-05  Product B    180         4   West
5  2023-01-06  Product A    210         5  North
6  2023-01-07  Product C    320         7   East
7  2023-01-08  Product B    160         3  South
8  2023-01-09  Product A    230         6  North
```

9	2023-01-10	Product C	310	7	East
10	2023-01-11	Product B	190	4	West
11	2023-01-12	Product A	240	6	North
12	2023-01-13	Product C	330	8	East
13	2023-01-14	Product B	170	3	South
14	2023-01-15	Product A	250	7	North
15	2023-01-16	Product C	340	8	East







RESULT:

Thus the code is executed successfully.